



OPEN Laboratory evaluation of *Beauveria bassiana* strain ATCC74040 as a potential biocontrol agent of *Aculops lycopersici*

Anna Sybilska¹, Nina Mądry¹, Elżbieta Wójcik-Gront² & Ewa Puchalska¹✉

The tomato russet mite (TRM), *Aculops lycopersici* (Tryon) (Eriophyidae) is a major pest of tomato crops worldwide, responsible for extensive plant damage and substantial yield losses. The limited availability of chemical control options for this pest, particularly in the European Union, highlights the need for alternative management strategies. The aim of this study was to assess the pathogenicity of the entomopathogenic fungus *Beauveria bassiana* strain ATCC74040 (in the form of biopesticide Naturalis®) against TRM females and eggs under laboratory conditions (25 ± 2 °C, 80% RH, 16:8 L/D), using a graded series of concentrations (10³–10⁷ CFU mL⁻¹). At the two highest levels tested (10⁶ and 10⁷ CFU mL⁻¹), direct spraying caused mean mortalities of 61.7% and 88.6% of TRM females after five days, respectively, whereas in the residual assay (exposure to plants treated 24 h earlier) the same concentrations resulted in only 8.2% and 16.4% mortality, correspondingly. The lethal concentrations required to cause 50% and 90% mortality of directly treated females within five days (LC₅₀ and LC₉₀) were 2.65 × 10⁵ CFU mL⁻¹ (95% CI 1.87 × 10⁵–3.74 × 10⁵) and 1.51 × 10⁷ CFU mL⁻¹ (95% CI 8.12 × 10⁶–2.83 × 10⁷), respectively. In contrast, in residual assays, LC₅₀ and LC₉₀ values were substantially higher—1.15 × 10⁹ (95% CI 1.41 × 10⁸–9.36 × 10⁹), and 7.81 × 10¹² CFU mL⁻¹ (95% CI 1.80 × 10¹¹–3.46 × 10¹⁴), in turn—highlighting the markedly lower efficacy of indirect application. At a concentration of 10⁶ CFU mL⁻¹, the fungus also exhibited ovicidal activity, reducing TRM hatching by average 42% compared to the control. The highest concentration tested (10⁷ CFU mL⁻¹) induced phytotoxic symptoms on tomato plants (cv. 'Mei Shuai') suggesting that cultivar-specific sensitivity may occur. Further research is needed to assess this risk across a wider range of tomato varieties, as this phenomenon could be critical for the practical use of Naturalis-based biocontrol strategies. The results indicate that *B. bassiana* can reduce TRM populations under direct application, but also underscore the limitations related to residual activity. These findings provide an essential basis for future semi-field and field trials to optimize application strategies and evaluate their integration into tomato crop protection programs.

Keywords Eriophyoid mites, *Beauveria bassiana*, Biological control, Entomopathogenic fungi

Tomato (*Solanum lycopersicum* L.) is one of the world's most widely consumed vegetables, with global production exceeding 186 million tons¹. Among its major pests is the tomato russet mite (TRM), *Aculops lycopersici* (Tryon) (Trombidiformes: Eriophyidae), the smallest known phytophagous mite infesting tomato crops. Originally described in Australia in 1917², TRM has since spread globally, becoming a significant eriophyoid pest^{3–5}. It primarily attacks Solanaceae species, including tomato, bell pepper (*Capsicum annuum* L.), petunia (*Petunia x hybrida* L.), potato (*Solanum tuberosum* L.) and weeds such as black nightshade (*Solanum nigrum* L.)⁶. Its minute size (150–200 µm), high fecundity, and short generation time make early detection difficult and can result in severe yield losses of 25–100%^{7,8}.

Chemical control of TRM with acaricides has proven to be ineffective and challenging likely due to the pest's ability to hide beneath glandular hairs on tomato leaves and stems, and its capacity for rapid resistance evolution under strong selection pressure^{8,9}. Active substances such as monocrotophos, pyridaphenthion, cyhexatin,

¹Department of Plant Protection, Institute of Horticultural Sciences, Warsaw University of Life Sciences—SGGW (WULS—SGGW), Nowoursynowska 159, 02-776 Warsaw, Poland. ²Department of Biometry, Institute of Agriculture, Warsaw University of Life Sciences—SGGW (WULS—SGGW), Nowoursynowska 159, 02-776 Warsaw, Poland. ✉email: ewa_puchalska@sggw.edu.pl

azocyclotin, chlorobenzilate, bromopropylate and dicofol have shown efficacy against TRM but have been banned for use in Europe⁸. Currently abamectin, milbemectin and pyridaben are registered for TRM control in Turkey but not in other EU countries^{10,11}. The lack of reliable and effective control methods for controlling the pest in this region, highlights a significant gap in the practical management of TRM.

While biological control represents an obvious and desirable alternative to chemical control approaches, no efficient or widely adopted method for managing TRM populations has been established to date. Several species of natural enemies, including *Amblyseius andersoni* (Chant), *Amblydromalus limonicus* (Garman and McGregor), *Amblyseius swirskii* (Athias-Henriot), *Neoseiulus californicus* (McGregor), *Neoseiulus cucumeris* (Oudemans), *Neoseiulus fallacis* (Garman), and *Typhlodromus (Anthoseius) recki* Wainstein, have been tested as biocontrol agents against this pest^{6,12}. However, these Phytoseiid mites are not used commercially, as the glandular hairs of tomato plants effectively hinder their reproduction and movement¹³. It has been observed that during TRM feeding, the trichomes become damaged and collapse, allowing predators to move. Nonetheless, by the time this occurs, the eriophyoids are already migrating to other parts of the plant where the glandular hairs are still intact, meaning that predators are never able to effectively reduce the pest population¹⁴. Recent studies have reported promising results for iolinid mites, such as *Homeopronematus anconai* (Baker) and *Pronematus ubiquitus* (McGregor) (Acari: Iolinidae)^{15,16}. Under laboratory and semi-field conditions, these species were shown to suppress TRM populations effectively; however, their performance still requires confirmation under commercial production conditions.

Entomopathogenic fungi (EPF) are also important agents in biological pest control¹⁷, infecting hosts through cuticle penetration and subsequent nutrient depletion, aided by hydrolytic enzymes. Initial infection involves spore adhesion mediated by surface proteins, such as hydrophobins and adhesins (MAD1 and MAD2), which facilitate host recognition^{18–20}. In addition to mechanical invasion, EPF produce toxic secondary metabolites—such as beauvericin, bassianolide, oosporein, and other compounds—that contribute to strain-specific virulence^{21–23}. The group of EPF most pathogenic to arthropods is the Entomophthorales, but the difficulties of conidial mass culture mean that species in this group are not used in the production of crop protection products²⁴. Currently, companies are producing biopreparations based on several different species of fungi belonging to the genera *Metarhizium*, *Beauveria*, *Paecilomyces*, *Isaria*, and *Lecanicillium* showing efficacy against different arthropods^{17,20,24,25}. One of the EPF species commonly used for biological pest control is *Beauveria bassiana* (Bals.-Criv.) Vuill., valued for its ease of mass production and widespread presence in the soil²⁶. It has been shown that *B. bassiana* colonizing the aboveground parts of about 25 plant species may increase their resistance to herbivorous arthropods^{20,27}, protect plants against some pathogens or improve their defense responses^{20,28,29}.

To date, only a few reports have addressed the infectivity of EPF against tomato russet mite (TRM), involving species such as *Hirsutella thompsonii* (Fisher)³⁰ and *Metarhizium anisopliae* (Metschnikoff) Sorokin³¹; however, none have advanced to practical application or integration into TRM management programs.

For this study, *B. bassiana* strain ATCC 74,040 was chosen because of its documented broad-spectrum activity, including efficacy against other mite species^{26,32,33}, and its availability as a commercial formulation (Naturalis®). These features make it a strong candidate for TRM control and could facilitate its rapid integration into Integrated Pest Management (IPM) programs upon validation. Our study evaluated its direct and residual activity against TRM females and the ovicidal effect on eriophyid mite eggs—a stage generally considered highly resistant to fungal infection.

Results

In this study, the statistical significance of the effects of two experimental factors—time after treatment and biopesticide concentration—on the mortality rate of TRM (dependent variable) were evaluated, based on a two-way ANOVA. The results showed that both time and concentration had significant effects on mortality, with *p*-values < 0.001 in both direct and residual application tests. However, no significant interaction between the two factors was observed (*p* = 0.602 for direct tests and *p* = 0.566 for residual tests), indicating that time and concentration independently influence TRM mortality rates without interactive effects.

Mortality of *A. lycopersici* females in the direct test

In direct application, at 1-day post-treatment, TRM mortality rates were low and ranged from average $0.8\% \pm 1\%$ (at treatment with 10^4 CFU mL⁻¹) to $9\% \pm 1\%$ (at 10^7 CFU mL⁻¹). According to the Dunn test, the average mortality induced by the spore concentration 10^7 CFU mL⁻¹ significantly differed from the lower concentrations (*p* = 0.004) (Fig. 1a). On the third day post-treatment, mortality rates increased, ranging from average of $3.9\% \pm 2\%$ (at 10^4 CFU mL⁻¹) to $31.1\% \pm 2\%$ (at 10^7 CFU mL⁻¹). The Dunn test showed that spore concentrations of 10^4 and 10^5 acted similarly (mean mortality $3.9\% \pm 2\%$ and $7.1\% \pm 2\%$ respectively), but 10^6 and 10^7 were significantly different from the lower concentrations and each other ($15.7\% \pm 2\%$ and $31.1\% \pm 2\%$ respectively; *p* < 0.001) (Fig. 1b). After five days, mortality further increased, ranging from $14.7\% \pm 2\%$ (at 10^4) to over $88\% \pm 3\%$ (at 10^7). The Dunn test revealed significant differences between all concentrations, with average mortality rates increasing with concentration (*p* < 0.001). The concentration 10^5 CFU mL⁻¹ recommended by the manufacturer for control of two spotted spider mite (*Tetranychus urticae* Koch) killed average $41\% \pm 2\%$ of TRM females (Fig. 1c). When using the highest tested spray concentration (10^7 CFU mL⁻¹) phytotoxicity of Naturalis® was observed on the leaves of tomato cultivar ‘Mei Shuai’ in the form of small spots with a diameter of 2–3 mm, within which the tissue was macerated.

Fungal growth was observed on $90.3\% \pm 3\%$ of the eriophyid mite cadavers transferred to PDA medium.

Mortality of *A. lycopersici* females in the residual test

In the residual test TRM females were placed on plants sprayed with biopesticide 24 h earlier. After one day of contact with the treated plant, TRM mortality rates were very low, and no significant differences were found

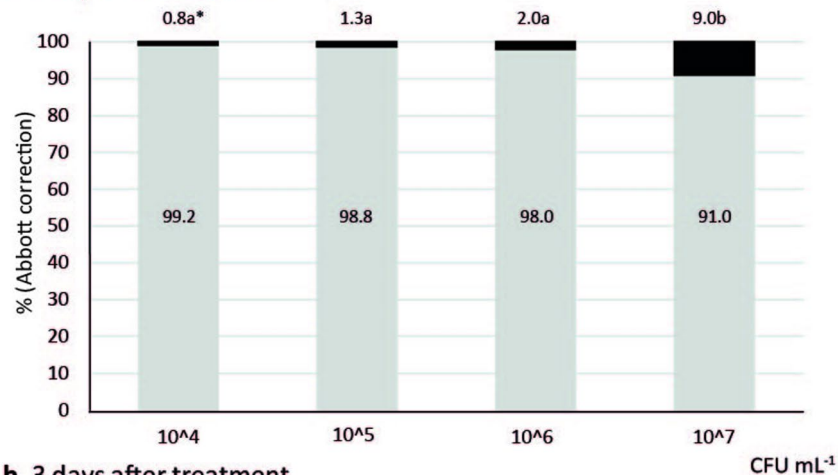
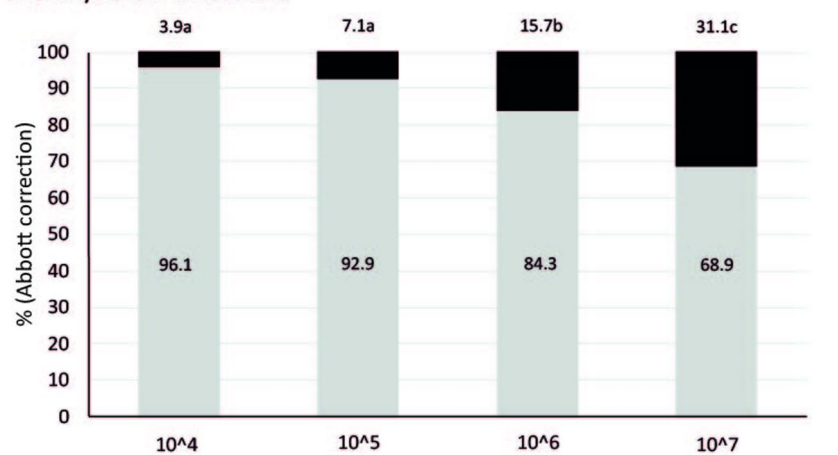
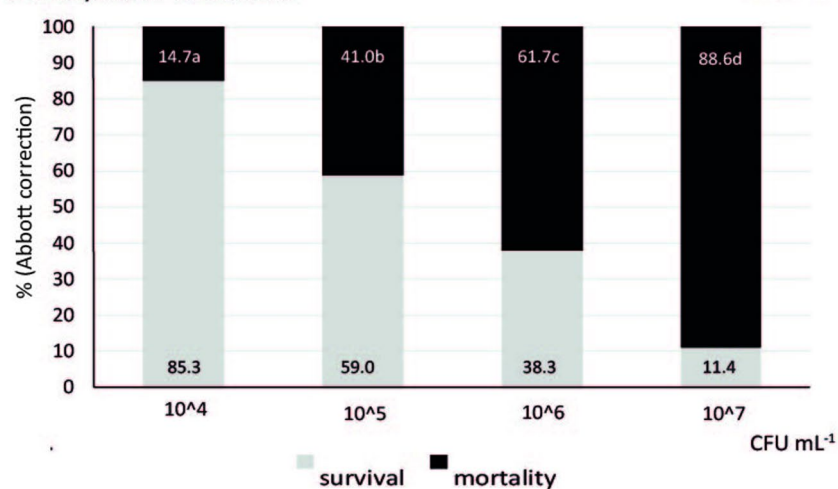
a. 1 day after treatment**b. 3 days after treatment****c. 5 days after treatment**

Fig. 1. (a–c) Survival/mortality of *Aculops lycopersici* females after direct treatment with *Beauveria bassiana* strain ATCC74040. For each time point (one, three, and five days), the Kruskal–Wallis test was used to compare the mortality rates across different fungal concentrations. According to Dunn's test, at a significance level of 0.05, the same letters indicate homogenous groups.

between the tested concentrations ($p=0.855$) (Fig. 2a). Three days after contact with plants sprayed with *B. bassiana*, female average mortality remained low (<9% at the highest spore concentration used) and did not differ significantly between treatments ($p=0.051$) (Fig. 2b). Significant differences between concentrations were observed only on the fifth day, when the highest concentration (10^6 CFU mL⁻¹) caused significantly higher

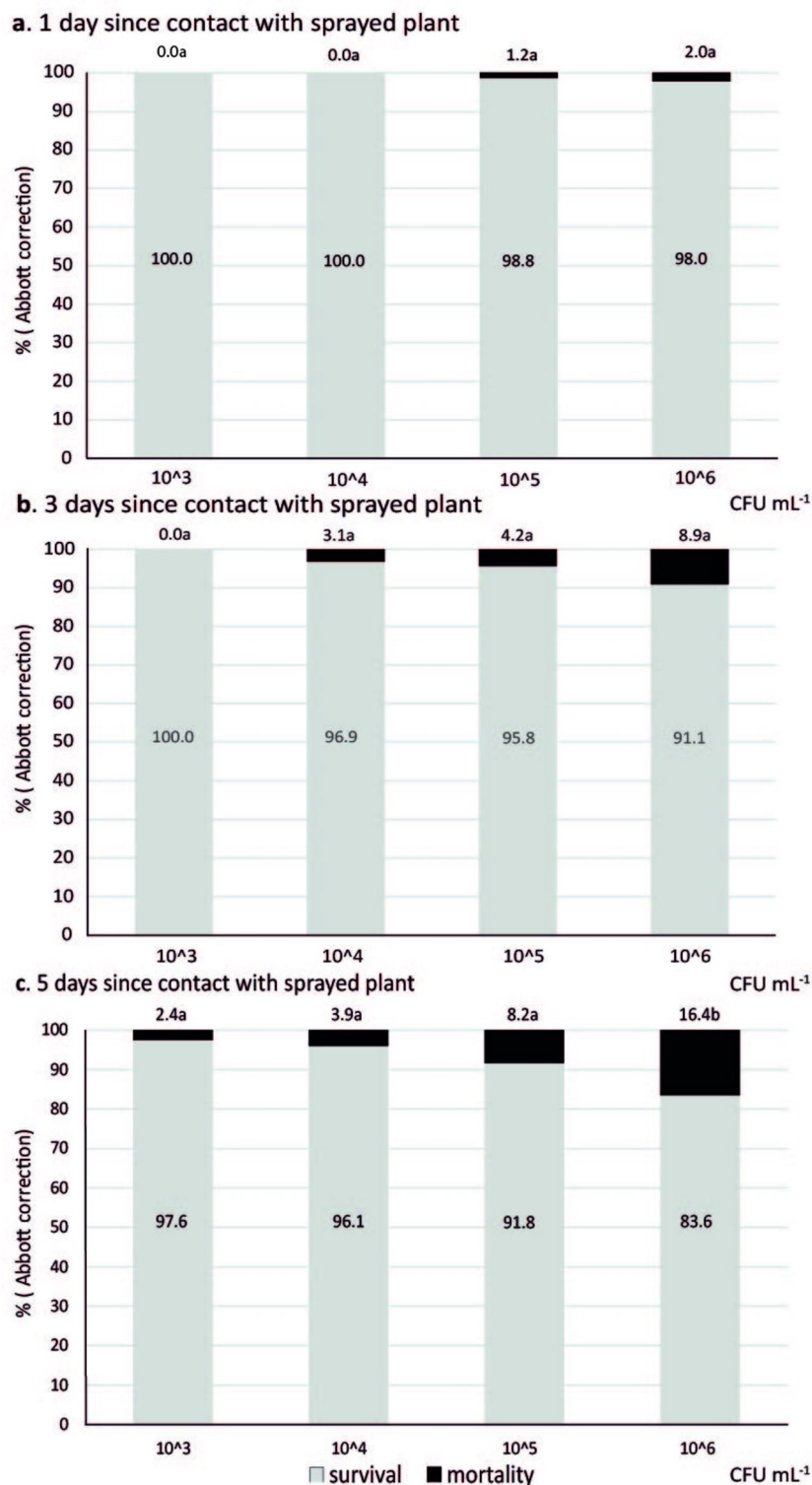


Fig. 2. (a–c) Residual effect of *Beauveria bassiana* strain ATCC74040 on survival/mortality of *Aculops lycopersici* females placed on plants 24 h after spraying with the biopesticide. The Kruskal–Wallis test was used to compare the mortality rates across different fungal concentrations. According to Dunn's test, at a significance level of 0.05, the same letters indicate homogenous groups.

mortality ($16.4\% \pm 2\%$) than the lower concentrations ($8.2\% \pm 1\%$ at 10^5 ; $3.9\% \pm 2\%$ at 10^4 ; $2.4\% \pm 1\%$ at 10^3 ; $p < 0.001$). Still the effect was very weak and mean over 80% of females remained alive (Fig. 2c).

Aculops lycopersici egg hatching after Naturalis® application

Spraying TRM eggs with *B. bassiana* at concentrations 10^3 , 10^4 and 10^5 CFU mL⁻¹ resulted in hatching rates comparable to those observed in the control. A significantly lower number of larvae was recorded when a concentration of 10^6 CFU mL⁻¹ was used, at which mean $42\% \pm 2\%$ of the eggs failed to hatch ($p < 0.001$) (Fig. 3). In all treatment groups, the highest number of larvae was observed on the third and fourth day after treatment (Fig. 4).

Beauveria bassiana strain ATCC74040 LC₅₀ and LC₉₀ values against *Aculops lycopersici*

In Table 1, all p -values are close to 1, suggesting a very good fit between the Probit model and the observed mortality data. It should be noted that the extremely high LC₅₀ and LC₉₀ values estimated for one and three days after treatment reflect the slow-acting nature of EPF and the fact that achieving 50% or 90% mortality within such a short period would theoretically require unrealistically high concentrations. This pattern is typical for pathogens that require time to infect and kill their hosts. By the fifth day post-treatment, the LC values dropped to biologically meaningful levels, particularly in the direct application test, which better reflects the practical potential of *B. bassiana* for TRM control. To achieve 50% mortality (LC₅₀) of treated individuals five days after direct spraying, a *B. bassiana* concentration of 2.65×10^5 CFU mL⁻¹ would be required. For 90% mortality (LC₉₀), the necessary concentration would be 1.51×10^7 CFU mL⁻¹. In contrast, the residual effect of the pathogen was weak. To reach the same mortality levels within five days after residual application, much higher concentrations— 1.15×10^9 CFU mL⁻¹ for LC₅₀ and 7.81×10^{12} CFU mL⁻¹ for LC₉₀—would be needed (Table 1).

Lethal time to kill *A. lycopersici* females treated with *B. bassiana* at the dosage of 10^5 CFU mL⁻¹ was estimated using Probit analysis (Fig. 5). The R^2 (coefficient of determination) value was 0.914, indicating that 91.4% of the variability in the Probit values is explained by the linear relationship with Log(T). This suggests a very good fit of the regression model to the data. This is confirmed by χ^2 goodness of fit analysis, where the p -value was 0.953. From the analysis, the LT₅₀ and LT₉₀ values were 6.5 and 16.9 days.

Discussion

Although EPF are commonly used mainly to control harmful insects^{34–36}, mites can also be good hosts for these pathogens^{24,37}. Their soft-bodied structure may facilitate the growth and development of mycelium²⁴, as confirmed by our study. Research on the control of herbivorous mites using EPF focuses mainly on spider mites (Tetranychidae)³⁷. These studies concern especially the natural incidence of EPF on tetranychid mites³⁸, the susceptibility of spider mites to fungal infections, and the effectiveness of EPF in spider mite management^{24,39–42}. Some EPF species, from the genera *Hirsutiella*, *Lecanicillium*, *Fusarium* or *Beauveria*, have also been obtained from cadavers of eriophyid mites (Eriophyidae)^{43,44} or from plant organs (fruits, leaves) infested by eriophyids⁴⁵. The efficacy of selected EPF isolates against eriophyid mites has been reported across diverse climatic regions, including tropical, subtropical, and temperate zones, for species such as: *Aceria guerreronis* Keifer^{43,45–47}, *Eriophyes prosopidis* Saxena⁴⁸, *Aculops cannabicola* (Farkas)⁴⁹, *Phyllocoptruta oleivora* Ashmead^{50,51}, *Phyllocoptes gracilis* Nalepa⁵² and *Phytoptus avellanae* Nalepa⁴⁴. Recently, reports from Africa on the use of *B. bassiana* for the control of TRM have also been published⁵³. Interestingly, substantial differences were observed between the current results and those obtained in Africa. A concentration of 2.3×10^4 CFU mL⁻¹ reduced the African

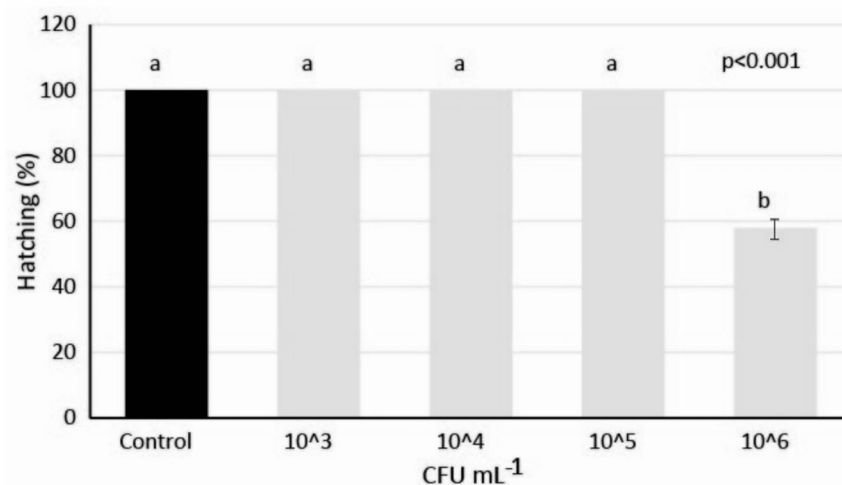


Fig. 3. Effect of direct spraying with different *Beauveria bassiana* strain ATCC74040 concentrations (color grey) on *Aculops lycopersici* egg hatching. According to Dunn's test, at a significance level of 0.05, the same letters indicate homogenous groups.

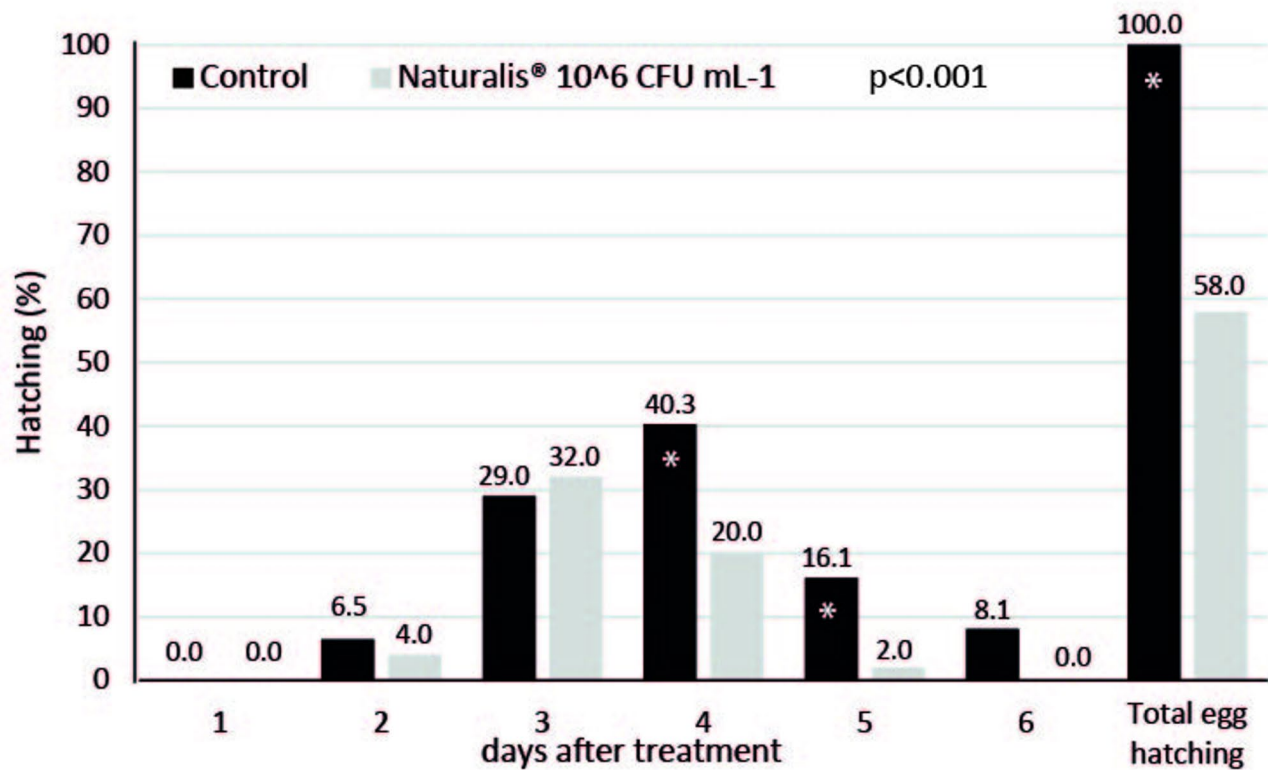


Fig. 4. *Aculops lycopersici* egg hatching in the following days after treatment *Beauveria bassiana* strain ATCC74040 at concentration 10⁶ CFU mL⁻¹. Asterisks indicate statistically significant differences between control and experimental groups ($p < 0.05$), based on Dunn’s test.

The pest exposure to biopesticide	Effect expected after (days) since treatment	a	b	χ ²	p-value	LC ₅₀ CFU mL ⁻¹	Lower 95% CI	Upper 95% CI	LC ₉₀ CFU mL ⁻¹	Lower 95% CI	Upper 95% CI
Direct	1	0.335	−3.849	0.050	0.997	3.14 × 10 ¹¹	1.25 × 10 ⁸	7.87 × 10 ¹⁴	2.09 × 10 ¹⁵	6.55 × 10 ⁹	6.85 × 10 ²⁰
	3	0.427	−3.532	0.011	1.000	1.84 × 10 ⁸	6.12 × 10 ⁷	5.54 × 10 ⁸	1.82 × 10 ¹¹	2.10 × 10 ¹⁰	1.61 × 10 ¹²
	5	0.729	−3.953	0.018	0.999	2.65 × 10 ⁵	1.87 × 10 ⁵	3.74 × 10 ⁵	1.51 × 10 ⁷	8.12 × 10 ⁶	2.83 × 10 ⁷
Residual	1	0.188	−3.180	0.000	1.000	8.80 × 10 ¹⁶	3.47 × 10 ¹⁴	2.23 × 10 ¹⁹	5.83 × 10 ²³	2.95 × 10 ²⁰	1.15 × 10 ²⁷
	3	0.257	−2.932	0.006	1.000	2.52 × 10 ¹¹	1.33 × 10 ⁹	4.76 × 10 ¹³	2.40 × 10 ¹⁶	1.01 × 10 ¹³	5.71 × 10 ¹⁹
	5	0.334	−3.027	0.007	1.000	1.15 × 10 ⁹	1.41 × 10 ⁸	9.36 × 10 ⁹	7.81 × 10 ¹²	1.80 × 10 ¹¹	3.46 × 10 ¹⁴

Table 1. Results of a Probit analysis conducted for both direct and residual tests at different time points (1, 3, and 5 days) to estimate the LC₅₀ and LC₉₀ values with 95% confidence intervals. Probit model parameters (a—slope and b—intercept), the Chi-square statistic (χ²), and the associated p-values.

eriphyoid mite population by 70% within five days⁵³, whereas a similar concentration applied to the European TRM population in the present research resulted in only ~15% mortality over the same period. These discrepancies may be attributed to differences in experimental conditions (controlled laboratory bioassays vs. commercial greenhouse trials). However, as both studies provided environmental conditions favorable for EPF infection and development, the observed variation may also reflect differences in pathogen–host specificity. It is known that different strains of EPF species, including *B. bassiana*, may exhibit varying specificity and virulence toward different host species^{54–56}. Moreover, even within the same host species, susceptibility to EPF can differ between geographically distinct populations^{55–57}. An example involving mites is provided by Perinotto et al.⁵⁸ who examined two geographically distinct tick populations of the *Rhipicephalus microplus* (Canestrini) and showed significant differences in mortality and hatching percentage when treated with *Metarhizium anisopliae* (Metsch.) Sorokin or *B. bassiana*. Thus, among the possible explanations for the observed differences in mortality between the European TRM population (this investigation) and the African population of the pest⁵³ is origin-related variation in host susceptibility to the ATCC74040 strain of *B. bassiana*. Experimental verification is still needed to confirm this possibility.

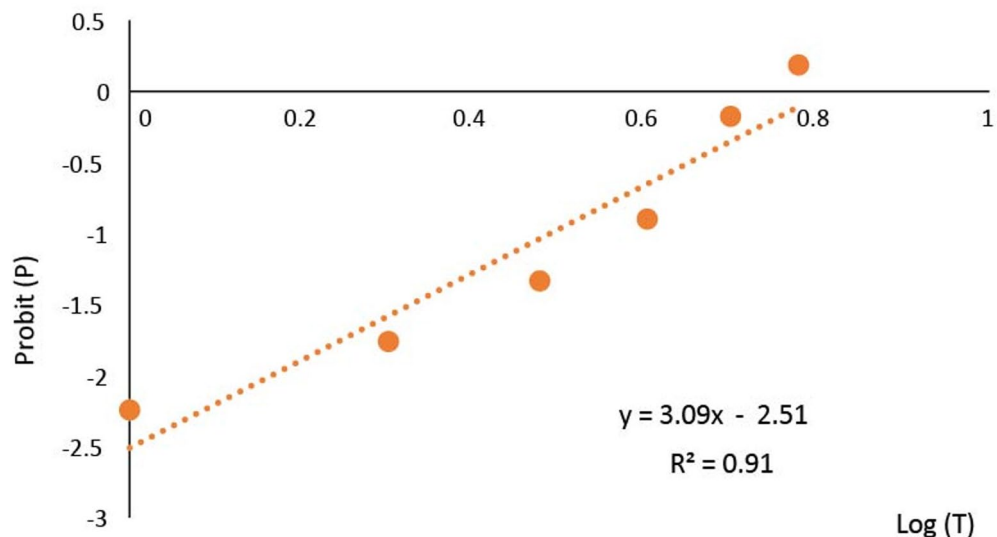


Fig. 5. A Probit analysis plot used to model the relationship between the logarithm of time (Log(T)) and the probability of *Aculops lycopersici* mortality (Probit(P)).

The direct application tests have shown that the standard dose of Naturalis® recommended for spider mites (e.g., *T. urticae*) (10^5 CFU mL⁻¹) is not sufficiently effective against TRM. At this concentration, 50% TRM mortality was observed after 6.5 days, while reaching 90% would require an average of 16.9 days. In comparison, applying a higher dose of *B. bassiana*— 1.51×10^7 CFU mL⁻¹—resulted in 90% pest mortality within five days, indicating that substantially higher concentrations are necessary to ensure rapid and effective control of TRM than of spider mites. However, this effective dose is comparable to—or even lower than—those reported for other eriophyid mite species. For instance, Minguely et al.⁵² reported 65% mortality of *P. gracilis* using *B. bassiana* at a concentration of 10^7 CFU mL⁻¹ under similar laboratory conditions. In turn, doses as high as 10^8 CFU mL⁻¹ were required to reduce *P. oleivora* by 44% in greenhouse trials⁵¹, and to achieve just 25% mortality of *A. guerrerensis* in laboratory tests⁴⁵. These comparisons suggest that both European (present study) and African⁵³ TRM populations exhibit relatively high susceptibility to the tested strain of *B. bassiana* compared to other eriophyid mites.

The above concentrations refer to the impact of *B. bassiana* on adults and/or juvenile mobile stages of eriophyid mites, but we also carried out tests to assess the ovicidal effect of this fungus. It is known that the developmental stage of the host plays an important role in the epizootics of EPF^{59,60}. In the case of mites, varying susceptibility to *B. bassiana* infection has been observed in larvae, nymphs, adults, and eggs of *T. urticae*⁶¹. Of all the developmental stages of herbivorous mites, eggs seem to be the most resistant to fungal infection. According to Sáenz-de-Cabezón Irigaray et al.⁶¹, the reason for the low susceptibility of spider mite eggs to fungal infections may be the unfavourable structure of the eggshell and/or the lack of an adequate amount of lipids enabling the germination and growth of EPF spores. In this study, concentrations from 10^3 to 10^5 CFU mL⁻¹ did not affect TRM hatching, but after the application of a dose of 10^6 CFU mL⁻¹ 42% of eggs were infected. This is the first report of EPF infectivity towards eriophyid mite eggs. From a plant protection perspective, the ability to suppress the egg stage may enhance the overall effectiveness of EPF-based biocontrol strategies, potentially leading to longer-lasting population suppression and reducing the need for repeated treatments.

The current study also demonstrated that the method of spores application affected the efficacy of *B. bassiana* against TRM. It was proved that residual application of spores is markedly less effective against TRM than direct application, with LC₅₀ values differing by over four orders of magnitude. This highlights the critical importance of application timing and method in EPF-based mite control. The germination of non-stressed *B. bassiana* spores and their penetration into the hemocoel typically take 18 to 24 h⁶². Therefore, if a host is not present within this period—as occurred in the residual test—the efficacy of EPF may be considerably diminished⁶³. Extended persistence of EPF spores on leaf surfaces increases their exposure to environmental factors that compromise viability⁶⁴. It has previously been shown that, in epigeal habitats, the persistence and virulence of EPF spores can be negatively affected by abiotic factors such as unfavorable temperatures (below 20 °C or above 35 °C), low relative humidity, or exposure to UV radiation⁶⁵. Among these, UV light is considered one of the main causes of EPF spores depletion and inactivation²⁵. In current study, chambers equipped with fluorescent lamps with a spectrum including UV-B light were used. Since exposure to UV-B radiation reduces the survival of *B. bassiana* spores over time⁶⁶, this could have contributed to the lower efficacy observed in the residual treatment, where spores applied to leaves were exposed to UV-B light for 24 h longer than in the direct spray test. Biotic factors, including the host or the plant on which the spores reside, may also influence spore viability. It has been shown that certain tomato alkaloids, such as tomatine, can inhibit the growth and development of *B. bassiana* spores⁶⁷. Therefore, the deposition of fungal spores on tomato leaves instead of directly on the host's body may have also contributed to the reduced efficacy of Naturalis® in the residual tests.

This study demonstrated the pathogenicity of the *B. bassiana* strain ATCC74040 towards TRM females and eggs, as confirmed by fungal outgrowth from cadavers, supporting its biocidal potential in pest control. However, we cannot rule out the possibility that the oil carrier in Naturalis® also exerts a biocidal effect on TRM. Unfortunately, the manufacturer does not disclose this ingredient, so we were unable to test its independent impact. Gatarayih et al.⁶⁸ investigated the effect of various adjuvants added to *B. bassiana* suspensions against spider mites. They showed that mineral oil (paraffin) alone had no acaricidal effect, causing spider mite mortality comparable to that observed after spraying with pure water (max. 2.5% dead individuals). Some mineral oils (*n*C21 Lovis and *n*C23 D-C-Tron NR) have also been tested against TRM. In laboratory experiments, strong pest population suppression was observed at 35 °C; however, the efficacy of the oils declined markedly at 25 °C⁶⁹. Since our experiments were conducted at 25 °C, it is unlikely that any of the oils tested in the aforementioned studies would have exerted a biocidal effect on TRM, even if they were components of Naturalis®. Nevertheless, it is worth noting that the highest tested concentration of Naturalis® (10^7 CFU mL⁻¹) caused phytotoxic effects on the leaves of the tomato cultivar ‘Mei Shuai’. This may have resulted from one of the formulation’s carrier components dissolving the wax layer on the leaf surface at such a high concentration, leading to tissue damage. Previous studies have shown that certain surfactants—such as the nonionic ESM20, anionic SDS, and cationic CTAB—can disrupt and solubilize the long-chain hydrocarbons and ester-based waxes that protect leaf surfaces, resulting in phytotoxicity^{70,71}. Since no reports of Naturalis®-induced phytotoxicity on tomatoes or other crops are currently available, this may represent a cultivar-specific response. Still, the finding is of practical relevance for crop protection and suggests that a small-scale trial of the biopesticide should be conducted before broader application in commercial cultivation.

In conclusion, this study provides baseline evidence that *B. bassiana* strain ATCC74040, formulated as Naturalis®, has potential as a component of tomato russet mite management, particularly in regions where chemical control options are limited. While residual activity was low and phytotoxicity at high concentrations warrants caution, these limitations underscore the importance of optimizing application method, spore concentration, and treatment timing within an IPM framework. Further greenhouse and field studies are essential to validate these findings, assess cultivar-specific sensitivity, and refine strategies for practical use under variable environmental conditions.

Methods

Plant material and mite colony

The ‘Mei Shuai’ tomato variety was used in the study. Seeds purchased from Seminis (Bayer Vegetables Poland) were sown into individual pots of garden soil mixed with perlite. When the plants had four compound leaves each, the leaves were detached and used in experiments.

Aculops lycopersici individuals were initially collected from commercial tomato crops under cover, across various regions of Poland. The stock colony was supplied with new individuals over a period of three years. Mass rearing of the pest was conducted on potted tomato ‘Mei Shuai’ plants in the Department of Plant Protection (Institute of Horticultural Sciences, Warsaw University of Life Sciences, Poland). Infested plants were kept in SANYO (MLR-350H) rearing chamber under controlled conditions (25 ± 2 °C temp., 70 ± 10% RH, 16:8 L/D photoperiod).

Fungal biocontrol agent

The bioinsecticide used was a commercial product Naturalis® (CBC (Europe) S.r.l. Italy) containing spores of *B. bassiana* strain ATCC74040 (2.3×10^7 CFU mL⁻¹) in a form of an oil-based concentrate (permission MRiRW in Poland no. R—50/2017 data access 27.12.2017). The viability of the *B. bassiana* spores was validated by germination on potato dextrose agar medium (PDA) in ten replicates. The plates were held at 25 °C. After 24 h cover slips were applied over media for the counts of germinated spores. A spore was considered germinated when the germ tube lengths were two times the diameter of the spore in question⁷². A total of 500 spores were evaluated under a microscope at 400 × magnification. Percentage germination was determined as 96.2% ± 0.62%.

Experimental designs

Experimental unit

From tomato leaves, discs with a diameter of 2 cm were cut with a corkboard and individually placed bottom side up in Petri dishes (5 cm diameter) lined with wet cotton wool. The edges of the leaf blades were secured with moistened cotton wool, to maintain leaf turgor and prevent mites escaping. In all combinations experimental units were kept in growth chamber SANYO (MLR-350H) under controlled conditions (25 °C ± 2 °C temp., 70 ± 10% RH, 16:8 L/D photoperiod).

Mortality of *A. lycopersici* females in the direct test

Ten age-synchronized TRM females were placed in each experimental arena. Discs with females were sprayed with a hand sprayer (Ribo Auto Intellectual Airbrush, METU 30 model, RoHS, China), with a spore suspension of *B. bassiana* at concentrations of 10^4 , 10^5 , 10^6 and 10^7 CFU mL⁻¹. Spore concentrations were determined with a hemocytometer and adjusted with sterile distilled water. The control treatment was conducted using sterile distilled water. Female mortality was assessed daily for a period of six days. In each combination, 100 TRM females were used, including the control combination, enabled studying of 500 individuals. In plant protection practice, the effect of significant reduction in the pest population size is expected after three to five days from the application of a biopreparation consisting of microorganisms. Therefore, the mortality of the mites was assessed at one, three, and five days after spraying. In the analysis, data were expressed as decimal fractions (probability of mortality) adjusted using Abbott’s formula for the third and fifth days, as deaths occurred in the control group. Mites were considered dead when no movement occurred under the microscope, even after a

gentle stimulation with the dissecting needle. Dead individuals from particular treatments were transferred to Petri dishes with sterile PDA medium supplemented with antibacterial agents (streptomycin sulphate 0.08% and penicillin 0.03%) to encourage fungal growth and sporulation in order to confirm that death was due to infection by *B. bassiana*^{73,74}. Petri dishes were sealed with Parafilm to maintain high humidity and incubated at 25 °C for five days, after which a microscopic examination was performed (Supl. S1). As many as 102 mite cadavers were examined in total.

Mortality of A. lycopersici females in the residual test

Experimental arenas, without mites, were sprayed with spore suspension at the concentrations 10^3 , 10^4 , 10^5 , 10^6 CFU mL⁻¹. We abandoned the highest dose of the pathogen used in the direct test due to the observed phytotoxicity symptoms it caused and replaced it with a lower one. In the control treatment, arenas were sprayed with sterile distilled water. The leaf discs were then air-dried under the laminar flow cabinet for 20 min and placed in a growth chamber (25 ± 2 °C temp., 80% RH, 16:8 L/D photoperiod). TRM females were applied to experimental units 24 h after spraying. Female mortality was observed for the next six days. In each combination, 100 TRM females were used, including the control combination, allowing the study of 500 individuals in residual test. The probability of mortality was determined in a similar manner to the direct test, and the cause of mites mortality was confirmed by microscopic examination.

Determination of egg hatching after Naturalis® application

In order to evaluate ovicidal effect of *B. bassiana*, two females of TRM were placed in each experimental unit to lay eggs. Six experimental units were prepared for each combination. After two days females were removed and the number of eggs they laid was counted. Each unit with eggs was sprayed with hand sprayer. The same spore concentrations as those in the residual test were used (10^3 , 10^4 , 10^5 and 10^6 CFU mL⁻¹) and sterile distilled water was used in the control combination. Over the next seven days, the number of larvae that hatched from eggs was determined. In each combination, including the control, 50–70 eggs were received for a total of 302 eggs tested.

Data analysis

Statistical analysis was performed using the data analysis software system Statistica (TIBCO Software Inc. Statistica; Version 13; TIBCO Software Inc.: Palo Alto, CA, USA, 2017). To compare mite mortality rates across different fungal concentrations at each of the three time points (one, three, and five days post-treatment), the Kruskal–Wallis test⁷⁵ was applied. Dunn's test with Holm adjustment served as a post-hoc procedure for pairwise comparisons.

Probit analysis estimated the lethal concentration that kills 50% of the mites (LC₅₀) and the concentration corresponding to a 90% mortality (LC₉₀). The Probit regression model (*b*) intercept represents the point where the regression line crosses the y-axis in the Probit model, reflecting the base level of mortality without considering the concentration. The Probit regression line (*a*) slope indicates how mortality increases with increasing concentrations of the fungal treatment. To determine the lethal time of TRM population (LT₅₀ and LT₉₀) treated directly with LC₅₀ dosage a Probit analysis plot modeling the relationship between the logarithm of time (Log(T)) and the probability of mortality (Probit(P)) was used. A chi-square test was conducted to check whether the regression curve fits the data significantly. A lower LC₅₀ and LC₉₀ indicates a more effective treatment for killing mites at lower concentrations.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Conceptualization—E.P., A.S., investigation—A.S., N.M. methodology—E.P., A.S., data analysis—A.S., E.P., visualization—A.S., E.P., formal analysis and software—E.W.-G., writing—original draft—A.S., E.P., writing—review and editing—A.S., E.P.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to E.P.

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